

HOLODECK

Machines virtuelles pour les ingénieurs de Starfleet

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9. Certificat SSL interne (web + FTP)
10. Serveur FTP en SSL/TLS,
11. Authentifier les utilisateurs web via LDAP
12. VM cliente — Banc de test
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1. MACHINES VIRTUELLES

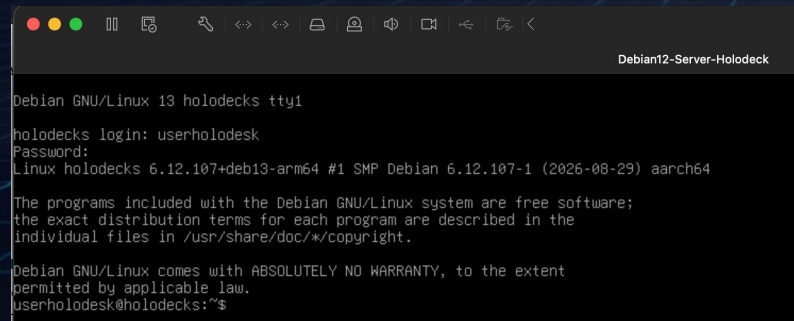
VM serveur (Debian CLI)

2 vCPU · 2 Go RAM · disque 32 Go ·

Cartes réseaux x 2 :

- WAN eth0 : NAT (DHCP Automatique)
- LAN eth1 : Host-Only (IP Statique : 192.168.100.1/24)

Paquet `open-vm-tools`



```
Debian12-Server-Holodeck

Debian GNU/Linux 13 holodecks tty1
holodecks login: userholodesk
Password:
Linux holodecks 6.12.107+deb13-arm64 #1 SMP Debian 6.12.107-1 (2026-08-29) aarch64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
userholodesk@holodecks:~$
```

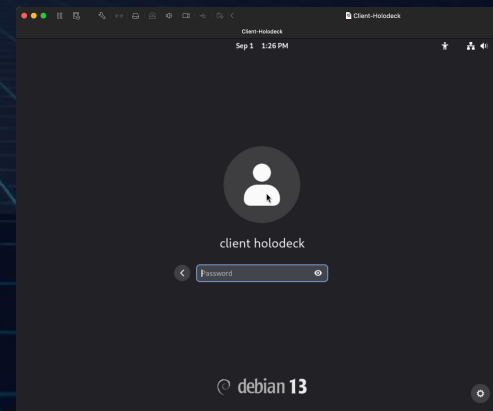
VM cliente (Debian GUI)

2 vCPU · 2 Go RAM · disque 16 Go

Carte réseau x 1:

- LAN Host-Only

Paquet `open-vm-tools-desktop`



Plan d'adressage

Domaine : starfleet.lan

Réseau LAN : 192.168.17.0/24

Serveur : 192.168.17.1 (statique)

Plage DHCP : .100 → .200 · bail 12 h

Noms : www8, www7, php, admin,

vscore, ftp, ldap → 192.168.17.1

ARCHITECTURE ET ADRESSAGE





```
root@enterprise:~# dpkg -l sudo  
dpkg-query: aucun paquet ne correspond à sudo
```

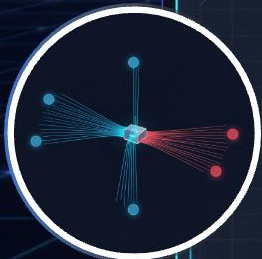
```
root@enterprise:~# sudo  
-bash: sudo : commande introuvable
```

2. PAS DE COMPTE SUDO

Principe de moindre privilège
Obligée à passer explicitement par root



- UFW sur dnsmasq
- Chaque règle liée à un service et à la carte LAN : **DHCP (67), DNS (53), SSH (22), Web (80, 443), FTP (21, 40000-40100), Cockpit (9090)**
- Rien d'ouvert côté WAN - administration depuis le LAN uniquement



- MariaDB, LDAP, Cockpit, code-server : en écoute sur 127.0.0.1, aucun port à ouvrir.
- Règle SSH temporaire d'installation retirée avant le rendu.
- Mise en place d'un service d'adressage et de nommage unifié avec Dnsmasq

3. PARE-FEU

```
root@enterprise:~# ufw status verbose
Status: active
Logging: on (low)
Default: deny (incoming), allow (outgoing), disabled (routed)
New profiles: skip
```

To	Action	From	
67/udp on ens37	ALLOW IN	Anywhere	# DHCP
53 on ens37	ALLOW IN	Anywhere	# DNS
22/tcp on ens37	ALLOW IN	Anywhere	# SSH
80,443/tcp on ens37	ALLOW IN	Anywhere	# HTTP HTTPS
21/tcp on ens37	ALLOW IN	Anywhere	# FTP
40000:40100/tcp on ens37	ALLOW IN	Anywhere	# FTP passif
67/udp (v6) on ens37	ALLOW IN	Anywhere (v6)	# DHCP
53 (v6) on ens37	ALLOW IN	Anywhere (v6)	# DNS
22/tcp (v6) on ens37	ALLOW IN	Anywhere (v6)	# SSH
80,443/tcp (v6) on ens37	ALLOW IN	Anywhere (v6)	# HTTP HTTPS
21/tcp (v6) on ens37	ALLOW IN	Anywhere (v6)	# FTP
40000:40100/tcp (v6) on ens37	ALLOW IN	Anywhere (v6)	# FTP passif

UFW / DNSmasq



```
root@enterprise:~# apt install -y ufw
ufw default deny incoming
ufw default allow outgoing
ufw allow in on ens37 to any port 67 proto udp comment 'DHCP'
ufw allow in on ens37 to any port 53 comment 'DNS'
ufw allow in on ens37 to any port 22 proto tcp comment 'SSH'
ufw allow in on ens33 to any port 22 proto tcp comment 'SSH admin PC - TEMPORAIRE retirer avant rendu'
ufw allow in on ens37 to any port 80,443 proto tcp comment 'HTTP HTTPS'
ufw allow in on ens37 to any port 21 proto tcp comment 'FTP'
ufw allow in on ens37 to any port 40000:40100 proto tcp comment 'FTP passif'
ufw logging low
ufw enable
```

```
apt install -y ufw
créer le fichier /etc/dnsmasq.d/starfleet.conf
```

```
root@enterprise: ~  
GNU nano 2.8 /etc/dnsmasq.d/starfleet.conf  
local=/starfleet.lan/  
expand-hosts  
domain-needed  
bogus-priv  
no-resolv  
server=0.0.0.0  
server=1.1.1.1  
  
host-record=enterprise.starfleet.lan,192.168.17.1  
cname=www7.starfleet.lan,enterprise.starfleet.lan  
cname=www8.starfleet.lan,enterprise.starfleet.lan  
cname=php.starfleet.lan,enterprise.starfleet.lan  
cname=admn.starfleet.lan,enterprise.starfleet.lan  
cname=score.starfleet.lan,enterprise.starfleet.lan  
cname=ftp.starfleet.lan,enterprise.starfleet.lan  
cname=leap.starfleet.lan,enterprise.starfleet.lan  
  
dhcp-authoritative  
dhcp-range=192.168.17.100,192.168.17.200,255.255.0,12h  
dhcp-option=option:router,192.168.17.1  
dhcp-option=option:dns-server,192.168.17.1  
dhcp-option=option:domain-name,starfleet.lan  
  
log-dhcp  
log-facility=/var/log/dnsmasq.log
```



Durcissement Système & Services
d'Infrastructure Réseau

3. PARE-FEU

Cette étape sert à dire au serveur DHCP sur quelle carte réseau il doit travailler.

- On modifie le fichier `/etc/default/isc-dhcp-server`

```
# Additional options to start dhcpd with.
# Don't use options -cf or -pf here; use DHCPD_CONF/ DHCPD_PID instead
#OPTIONS=""

# On what interfaces should the DHCP server (dhcpd) serve DHCP requests?
# Separate multiple interfaces with spaces, e.g. "eth0 eth1".
INTERFACESv4="ens37"
INTERFACESv6=""
```

```
subnet 192.168.50.0 netmask 255.255.255.0 {
    range 192.168.50.100 192.168.50.200;
    option routers 192.168.50.1;
    option domain-name-servers 192.168.50.1;
    option domain-name "starfleet.lan";
    default-lease-time 600;
    max-lease-time 7200;
}
```

Cette configuration permet au serveur de savoir précisément quoi donner aux machines qui se connectent :

Domaine / Timer / Adresse concerné

```
root@Server:~# systemctl restart isc-dhcp-server
root@Server:~# systemctl enable isc-dhcp-server
isc-dhcp-server.service is not a native service, redirecting to systemd-sysv-install.
Executing: /lib/systemd/systemd-sysv-install enable isc-dhcp-server
root@Server:~# systemctl status isc-dhcp-server
• isc-dhcp-server.service - LSB: DHCP server
   Loaded: loaded (/etc/init.d/isc-dhcp-server; generated)
   Active: active (running) since Tue 2026-09-08 14:08:02 CEST; 5min ago
     Docs: man:systemd-sysv-generator(8)
    Tasks: 1 (limit: 2256)
   Memory: 4.9M
      CPU: 43ms
   CGroup: /system.slice/isc-dhcp-server.service
           └─1583 /usr/sbin/dhcpd -4 -q -cf /etc/dhcp/dhcpd.conf ens37

sept. 08 14:08:00 Server systemd[1]: Starting isc-dhcp-server.service - LSB: DHCP server...
sept. 08 14:08:00 Server isc-dhcp-server[1571]: Launching IPv4 server only.
sept. 08 14:08:00 Server dhcpd[1583]: Wrote 0 leases to leases file.
sept. 08 14:08:00 Server dhcpd[1583]: Server starting service.
sept. 08 14:08:02 Server isc-dhcp-server[1571]: Starting ISC DHCPv4 server: dhcpd.
sept. 08 14:08:02 Server systemd[1]: Started isc-dhcp-server.service - LSB: DHCP server.
root@Server:~#
```

4. SERVEUR DHCP

isc-dhcp-server

```
systemctl restart isc-dhcp-server
systemctl enable isc-dhcp-server
systemctl status isc-dhcp-server # affiche 'active (running)'
```

bind9 / Zone starfleet.lan

- On déclare le nom de domaine local
- On dit que notre serveur est **le serveur principal**
- On indique où se trouve le fichier qui contient les vraies données de la zone

```
// Do any local configuration here
//
// Consider adding the 1918 zones here, if they are not used in your
// organization
//include "/etc/bind/zones.rfc1918";
zone "starfleet.lan" {
    type master;
    file "/etc/bind/zones/db.starfleet.lan";
};
```

5. SERVEUR DNS

```
GNU nano 7.2 /etc/bind/zones/db.starfleet.lan
$TTL 604800
@ IN SOA ns.starfleet.lan. admin.starfleet.lan. (
2024090301 ; serial (AAAAAMJJnn - a incrementer a CHAQUE modification)
604800 ; refresh
86400 ; retry
2419200 ; expire
604800 ) ; TTL negatif

@ IN NS ns.starfleet.lan.
ns IN A 192.168.50.1
www0 IN A 192.168.50.1
www7 IN A 192.168.50.1
php IN A 192.168.50.1
admin IN A 192.168.50.1
vscore IN A 192.168.50.1
```

- Durée en seconde /7D
- (Start Of Authority) → c'est la "carte d'identité" de la zon
- Le nom du serveur DNS principal

Ce fichier contient les données réelles de notre zone DNS. Le bloc SOA donne les informations administratives de la zone, avec un numéro de série à incrémenter à chaque modification.

-Vérification configuration :

Named-checkconf: qui contrôle la syntaxe générale

Named-checkzone: qui vérifie spécifiquement notre fichier de zone.

```
root@Server:~# named-checkconf
root@Server:~# named-checkzone starfleet.lan /etc/bind/zones/db.starfleet.lan
zone starfleet.lan/IN: loaded serial 2024090301
OK
root@Server:~#
```

```
root@Server:~# systemctl enable named
Synchronizing state of named.service with SysV service script with /lib/systemd/systemd-sysv-install
Executing: /lib/systemd/systemd-sysv-install enable named
root@Server:~# systemctl start named
root@Server:~# systemctl status named
● named.service - BIND Domain Name Server
   Loaded: loaded (/lib/systemd/system/named.service; enabled; preset: enabled)
   Active: active (running) since Thu 2026-09-10 09:18:01 CEST; 2h 3min ago
     Docs: man:named(8)
  Main PID: 3015 (named)
    Status: "running"
     Tasks: 8 (limit: 2256)
    Memory: 50.2M
       CPU: 356ms
    CGroup: /system.slice/named.service
            └─3015 /usr/sbin/named -f -u bind
```

5. SERVEUR DNS

bind9
Zone starfleet.lan

6. ANNUAIRE LDAP

OpenLDAP

Configuration de slapd

Le nom de domaine DNS est utilisé pour établir le nom distinctif de base (« base DN » ou « D ») vous indiquez « toto.example.org » ici, le nom distinctif de base sera « dc=toto, dc=example »

Nom de domaine :

starfleet.lan

<Ok>

Structure unités organisationnelles

```
GNU nano 7.2
dn: ou=people,dc=starfleet,dc=lan
objectClass: organizationalUnit
ou: people

dn: ou=groups,dc=starfleet,dc=lan
objectClass: organizationalUnit
ou: groups_
```

```
_bind: invalid credentials (49)
t@Server:~# ldapadd -x -D "cn=admin,dc=starfleet,dc=lan" -W -f structure.ldif
Enter LDAP Password:
adding new entry "ou=people,dc=starfleet,dc=lan"

adding new entry "ou=groups,dc=starfleet,dc=lan"
```

6. ANNUAIRE LDAP

```
dn: uid=picard,ou=pepeople,dc=starfleet,dc=lan
objectClass: inetOrgPerson
objectClass: posixAccount
objectClass: shadowAccount
uid: picard
sn: Picard
givenName: Jean-Luc
cn: Jean-Luc Picard
displayName: Jean-Luc Picard
uidNumber: 10000
gidNumber: 10000
userPassword: {CRYPT}x
loginShell: /bin/bash
homeDirectory: /home/picard
```

```
root@Server:~# ldapadd -x -D "cn=admin,dc=starfleet,dc=lan" -W -f utilisateur.ldif
Enter LDAP Password:
adding new entry "uid=picard,ou=people,dc=starfleet,dc=lan"

root@Server:~# ldappasswd -x -D "cn=admin,dc=starfleet,dc=lan" -W -S "uid=picard,ou=people,dc=starfleet,dc=lan"
New password:
Re-enter new password:
Enter LDAP Password:
```

```
# extended LDIF
#
# LDAPv3
# base <dc=starfleet,dc=lan> with scope subtree
# filter: (objectClass=*)
# requesting: ALL
#
# starfleet.lan
dn: dc=starfleet,dc=lan
objectClass: top
objectClass: dcObject
objectClass: organization
o: Starfleet
dc: starfleet

# people, starfleet.lan
dn: ou=people,dc=starfleet,dc=lan
objectClass: organizationalUnit
ou: people

# groups, starfleet.lan
dn: ou=groups,dc=starfleet,dc=lan
objectClass: organizationalUnit
ou: groups

# picard, people, starfleet.lan
dn: uid=picard,ou=people,dc=starfleet,dc=lan
objectClass: inetOrgPerson
objectClass: posixAccount
objectClass: shadowAccount
uid: picard
sn: Picard
givenName: Jean-Luc
cn: Jean-Luc Picard
displayName: Jean-Luc Picard
uidNumber: 10000
gidNumber: 10000
loginShell: /bin/bash
homeDirectory: /home/picard

# search result
search: 2
result: 0 Success

# numResponses: 5
# numEntries: 4
```

```
you already have your root account protected, so you can safely answer 'n'.

Change the root password? [Y/n] Y
New password:
Re-enter new password:
Password updated successfully!
Reloading privilege tables..
... Success!

By default, a MariaDB installation has an anonymous user, allowing anyone
to log into MariaDB without having to have a user account created for
them. This is intended only for testing, and to make the installation
go a bit smoother. You should remove them before moving into a
production environment.

Remove anonymous users? [Y/n] Y
... Success!

Normally, root should only be allowed to connect from 'localhost'. This
ensures that someone cannot guess at the root password from the network.

Disallow root login remotely? [Y/n] Y
... Success!

By default, MariaDB comes with a database named 'test' that anyone can
access. This is also intended only for testing, and should be removed
before moving into a production environment.

Remove test database and access to it? [Y/n] Y
- Dropping test database...
... Success!
- Removing privileges on test database...
... Success!

Reloading the privilege tables will ensure that all changes made so far
will take effect immediately.

Reload privilege tables now? [Y/n] Y
... Success!

Cleaning up...

All done! If you've completed all of the above steps, your MariaDB
installation should now be secure.

Thanks for using MariaDB!
```

7. PHP 7/8 MARIADB NGINX

PHP 7.x et 8.x via le dépôt Sury

Dernières versions, hors dépôts Debian

Nginx - nginx.org

1.31.4 (mainline)

Debian : 1.26.3

MariaDB - dépôt officiel

12.3.3

Debian : 11.8.6

PHP - dépôt Sury

8.5.9 et 7.4.33

Debian : 8.4 uniquement

*PHP 8.6.0~alpha1 disponible mais écarté :
une pré-version n'est pas une « dernière version » exploitable.*

7. PHP 7/8 MARIADB NGINX

```
root@enterprise:~# apt install -y gnupg2 ca-certificates lsb-release apt-transport-https
curl -fsSL https://nginx.org/keys/nginx_signing.key | gpg --dearmor -o /usr/share/keyrings/nginx-archive-keyring.gpg
echo "deb [signed-by=/usr/share/keyrings/nginx-archive-keyring.gpg] http://nginx.org/packages/mainline/debian $(lsb_release -cs) nginx" > /etc/apt/sources.list.d/nginx.list
```

```
root@enterprise:~# cd /tmp && curl -LsSO https://r.mariadb.com/downloads/mariadb_repo_setup
bash mariadb_repo_setup
```

```
root@enterprise:~# curl -fsSL /tmp/sury.deb https://packages.sury.org/debsuryorg-archive-keyring.deb && dpkg -i /tmp/sury.deb
echo "deb [signed-by=/usr/share/keyrings/deb.sury.org-php.gpg] https://packages.sury.org/php/ $(lsb_release -cs) main" > /etc/apt/sources.list.d/php.list
```

```
root@enterprise:~# apt update
apt-cache policy nginx mariadb-server | grep -A1 Candidat
apt-cache search '^php8\.[0-9]-fpm$'
apt-cache search '^php7\.[0-9]-fpm$'
```

```
root@enterprise:~# V=8.5
apt install -y nginx mariadb-server \
  php$V-fpm php$V-mysql php$V-ldap php$V-mbstring php$V-xml php$V-zip php$V-curl php$V-gd \
  php7.4-fpm php7.4-mysql php7.4-mbstring php7.4-xml
```

www7 → php7.4-fpm.sock

www8 → php8.5-fpm.sock

7. PHP 7/8 MARIADB NGINX

Deux services PHP-FPM, deux sockets unix.
Chaque vhost Nginx choisit son socket : même code, deux moteurs.

```
root@enterprise:~# nginx -v
php8.5 -v | head -1
php7.4 -v | head -1
mariadb --version
systemctl is-active nginx php8.5-fpm php7.4-fpm mariadb
nginx version: nginx/1.31.4
PHP 8.5.9 (cli) (built: Jul 30 2026 14:58:20) (NTS)
PHP 7.4.33 (cli) (built: Jul 3 2026 10:06:58) ( NTS )
mariadb from 12.3.3-MariaDB, client 15.2 for debian-linux-gnu (x86_64) using EditLine wrapper
inactive
active
active
active
```

7. NGINX : VHOSTS / HTTPS

Accès MariaDB via PhpMyAdmin

- Port 80 : redirection 301 vers HTTPS - aucun contenu en clair
- TLS 1.2/1.3, HSTS activé
- `php.starfleet.lan` : phpMyAdmin (dernière version, phpmyadmin.net)
- MariaDB en écoute locale seulement

```
root@enterprise:~# cat > /etc/nginx/ssl-params.conf <<'EOF'
ssl_certificate      /etc/ssl/starfleet/starfleet.crt;
ssl_certificate_key  /etc/ssl/starfleet/starfleet.key;
ssl_protocols TLSv1.2 TLSv1.3;
ssl_prefer_server_ciphers on;
add_header Strict-Transport-Security "max-age=31536000" always;
EOF
```

```
root@enterprise:~# cd /tmp
curl -LO https://www.phpmyadmin.net/downloads/phpMyAdmin-latest-all-languages.tar.gz
tar xzf phpMyAdmin-latest-all-languages.tar.gz
cp -r phpMyAdmin-*all-languages/. /var/www/php/
cp /var/www/php/config.sample.inc.php /var/www/php/config.inc.php
sed -i "s|['\"]blowfish_secret['\"]|['\"]blowfish_secret['\"] = '\$(openssl rand -base64 24)'|" /var/www/php/config.inc.php
grep blowfish /var/www/php/config.inc.php
chown -R www-data:www-data /var/www
ls /var/www /var/www/php | head
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           Dload  Upload   Total   Spent    Left   Speed
  0     0    0     0    0     0      0      0  --:--:-- --:--:-- --:--:--    0
100 13.7M 100 13.7M    0     0  9287k    0  0:00:01 0:00:01 --:--:-- 12.4M
$cfg['blowfish_secret'] = 'TT3Y00FoLKFapRtVqMpS4015wbtXzm6E'; /* YOU MUST FILL IN THIS FOR COOKIE AUTH! */
/var/www:
auth
php
www7
www8

/var/www/php:
babel.config.json
Changelog
composer.json
```

```
root@enterprise:~# cat > /etc/nginx/conf.d/00-redirect.conf <<'EOF'
server {
    listen 80 default_server;
    server_name _;
    return 301 https://$host$request_uri;
}
EOF
```

7. NGINX : VHOSTS / HTTPS

phpMyAdmin

Starfleet Engineering - PHP 8.5.9

MariaDB : 12.3.3-MariaDB-deb13

PHP Version 8.5.9

System	Linux enterprise 6.12.107+deb13-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.12.107-1 (2026-08-29) x86_64
Build Date	Jul 30 2026 14:58:20
Build System	Linux
Build Provider	Debian
Server API	FPM/FastCGI
Virtual Directory Support	disabled

Starfleet Engineering - PHP 7.4.33

MariaDB : 12.3.3-MariaDB-deb13

PHP Version 7.4.33

System	Linux enterprise 6.12.107+deb13-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.12.107-1 (2026-08-29) x86_64
Build Date	Jul 3 2026 10:06:58
Server API	FPM/FastCGI
Virtual Directory Support	disabled
Configuration File (php.ini) Path	/etc/php/7.4/fpm
Loaded Configuration File	/etc/php/7.4/fpm/php.ini
Scan this dir for additional .ini files	/etc/php/7.4/fpm/conf.d

phpMyAdmin

Bienvenue dans phpMyAdmin

Langue (Language)
Français - French

Connexion

Utilisateur : root

Mot de passe : *****

Connexion

Redirection :

Starfleet Engineering - PHP 8.5.9

MariaDB : 12.3.3-MariaDB-deb13

PHP Version 8.5.9

System	Linux enterprise 6.12.107+deb13-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.12.107-1 (2026-08-29) x86_64
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Starfleet Engineering - PHP 8.5.9

MariaDB : 12.3.3-MariaDB-deb13

PHP Version 8.5.9

System	Linux enterprise 6.12.107+deb13-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.12.107-1 (2026-08-29) x86_64
Build Date	Jul 30 2026 14:58:20
Build System	Linux
Build Provider	Debian
Server API	FPM/FastCGI
Virtual Directory Support	disabled

9. CERTIFICAT SSL INTERNE - WEB + FTP

```
root@enterprise:~/ca# cat > /root/ca/openssl.cnf <<'EOF'
[req]
prompt = no
distinguished_name = dn
req_extensions = v3_req
[dn]
C = FR
O = Starfleet
CN = starfleet.lan
[v3_req]
subjectAltName = @alt → SAN
[alt]
DNS.1 = starfleet.lan
DNS.2 = *.starfleet.lan
EOF
```

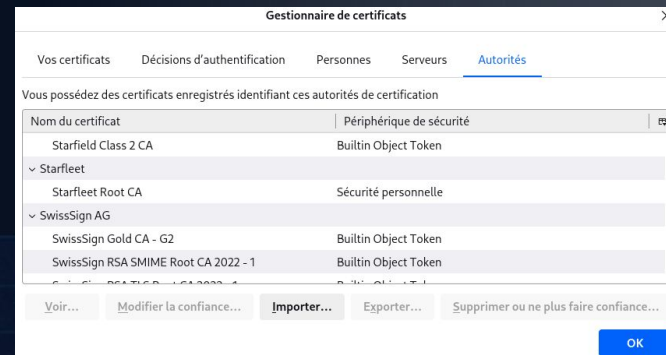
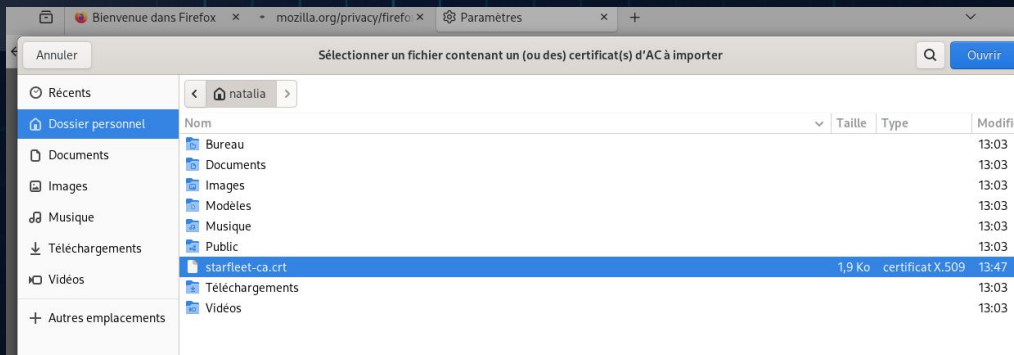
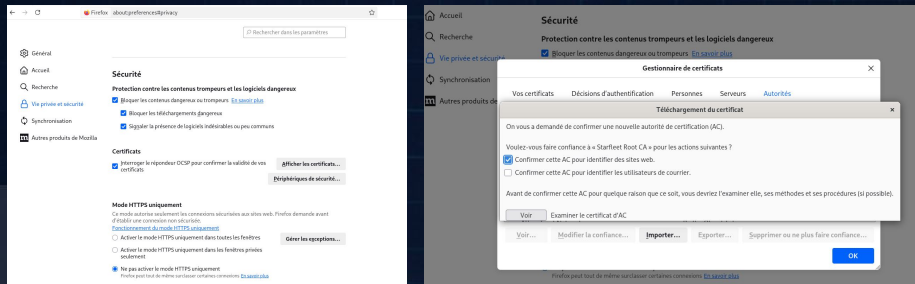
```
root@enterprise:~# mkdir -p /root/ca && cd /root/ca
openssl genrsa -out starfleet-ca.key 4096
openssl req -x509 -new -sha256 -days 3650 -key starfleet-ca.key -subj "/C=FR/O=Starfleet/CN=Starfleet Root CA" -out starfleet-ca.crt
ls -l
total 8
-rw-rw-r-- 1 root root 1919  1 sept. 13:37 starfleet-ca.crt
-rw-rw-r-- 1 root root 3268  1 sept. 13:37 starfleet-ca.key
```

- CA locale « Starfleet Root CA » (10 ans)
- Certificat serveur wildcard : SAN starfleet.lan + *.starfleet.lan
- Un seul certificat pour Nginx et vsftpd, comme demandé
- CA importée dans Firefox → cadenas fermé, aucun avertissement

```
tf file openssl.cnf -out starfleet.crt
mkdir -p /etc/ssl/starfleet
cp starfleet.crt starfleet-ca.crt /etc/ssl/starfleet/
cp starfleet.key /etc/ssl/starfleet/ && chmod 600 /etc/ssl/starfleet/starfleet.key
Certificate request self-signature ok
subject=C=FR, O=Starfleet, CN=starfleet.lan
```

```
root@enterprise:~/ca# openssl x509 -in /etc/ssl/starfleet/starfleet.crt -noout -subject -issuer -dates -ext subjectAltName
subject=C=FR, O=Starfleet, CN=starfleet.lan
issuer=C=FR, O=Starfleet, CN=Starfleet Root CA
notBefore=Sep  1 11:45:44 2026 GMT
notAfter=Dec  4 11:45:44 2028 GMT
X509v3 Subject Alternative Name:
    DNS:starfleet.lan, DNS:*.starfleet.lan
```

9. CERTIFICAT SSL INTERNE - WEB + FTP



10. SERVEUR FTP EN SSL/TLS

- vsftpd - FTPS explicite : AUTH TLS sur le port 21, TLS 1.3 négocié
- TLS obligatoire : force_local_logins_ssl (authentification) + force_local_data_ssl (données)
- Même certificat que le web
- chroot_local_user : webmaster enfermé dans /var/www
- Compte sans shell (nologin) - déclaré dans /etc/shells pour PAM
- Test : fichier déposé dans www8/ → servi en HTTPS immédiatement

Source :

<https://www.linuxtricks.fr/wiki/vsftpd-le-fichier-de-configuration-vsftpd-conf>

```
natalia@enterprise:~$ su -
Mot de passe :
root@enterprise:~# adduser --disabled-login --gecos "Webmaster Starfleet" webmaster
passwd webmaster
chown root:root /var/www && chmod 755 /var/www
chown -R webmaster:www-data /var/www/www7 /var/www/www8
chmod -R 2775 /var/www/www7 /var/www/www8
echo webmaster > /etc/vsftpd.userlist
Nouveau mot de passe :
Retapez le nouveau mot de passe :
passwd : mot de passe mis à jour avec succès
```

```
natalia@enterprise:~$ su -
Mot de passe :
root@enterprise:~# adduser --disabled-login --gecos "Webmaster Starfleet" webmaster
passwd webmaster
chown root:root /var/www && chmod 755 /var/www
chown -R webmaster:www-data /var/www/www7 /var/www/www8
chmod -R 2775 /var/www/www7 /var/www/www8
echo webmaster > /etc/vsftpd.userlist
Nouveau mot de passe :
Retapez le nouveau mot de passe :
passwd : mot de passe mis à jour avec succès
root@enterprise:~# apt install -y vsftpd
E: /etc/vsftpd.conf: /etc/vsftpd.conf.orig
cat > /etc/vsftpd.conf <<EOF
listen=YES
listen_ipaddr=
anonymous_enable=NO
local_enable=YES
write_enable=YES
local_umask=022
dirmessage_enable=YES
xferlog_enable=YES

# Chroot sur le dossier web
chroot_local_user=YES
local_root=/var/www

# Utilisateurs autorisés
userlist_enable=YES
userlist_file=/etc/vsftpd.userlist
userlist_deny=NO

# TLS avec le certificat Starfleet
ssl_enable=YES
force_local_logins_ssl=YES
force_local_data_ssl=YES
ssl_tlsv1=YES
ssl_tlsv1_2=YES
ssl_tlsv1_3=YES
ssl_sslv2=NO
ssl_sslv3=NO
require_ssl_reuse=NO
rsa_cert_file=/etc/ssl/starfleet/starfleet.crt
rsa_private_key_file=/etc/ssl/starfleet/starfleet.key

# Mode passif (ports ouverts dans ufw)
passv_enable=YES
passv_min_port=8080
passv_max_port=8100
passv_address=192.168.17.1
EOF
systemctl restart vsftpd
systemctl is-active vsftpd
```

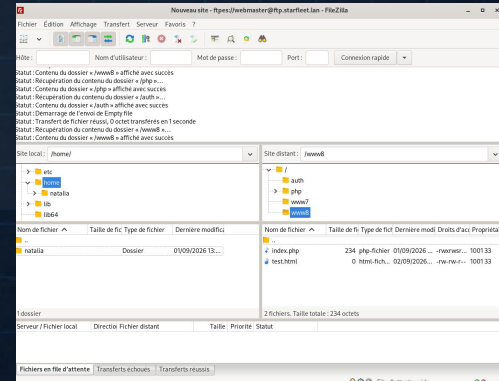
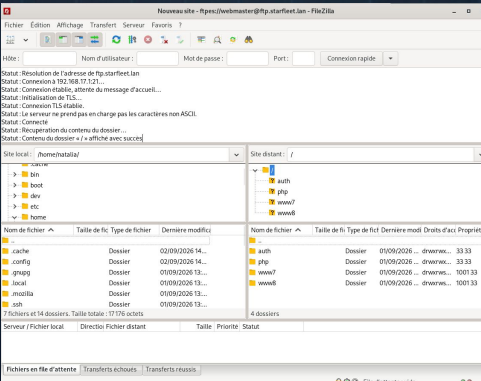
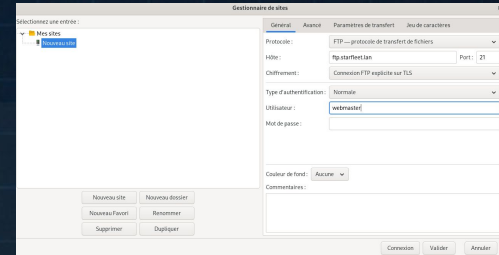
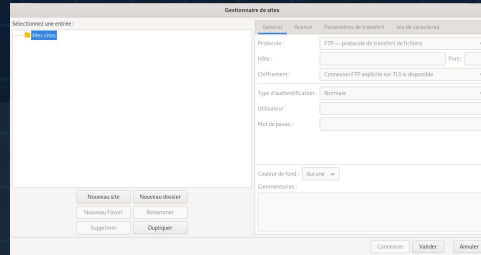
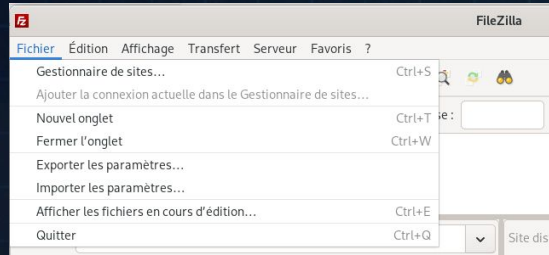
→ Enferme l'utilisateur dans son dossier

→ Active le chiffrement

→ Chiffrement obligatoire

→ Certificat et clé (mêmes que le web)

10. SERVEUR FTP EN SSL/TLS

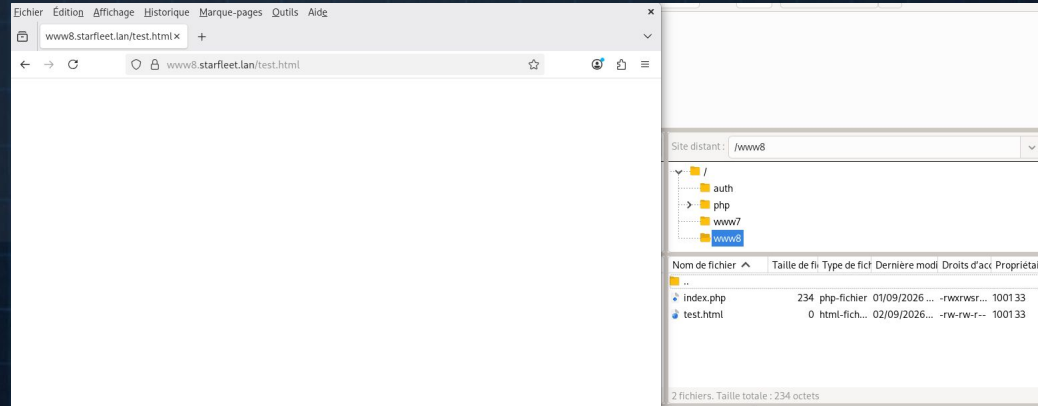


10. SERVEUR FTP EN SSL/TLS

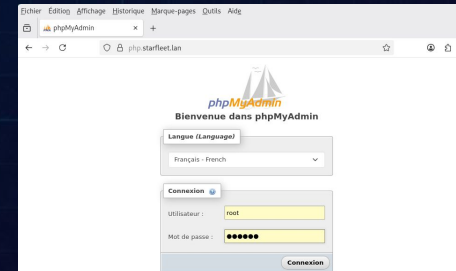
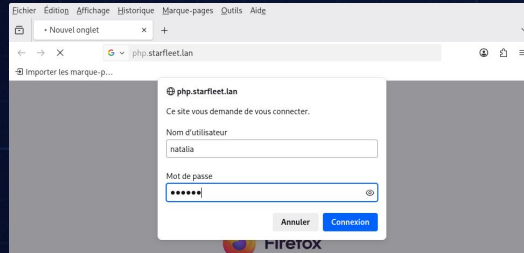
```
root@enterprise:~# mkdir -p /home/natalia/export/vsftpd  
cp /etc/vsftpd.conf /etc/vsftpd.userlist /home/natalia/export/vsftpd/  
chown -R natalia:natalia /home/natalia/export
```

```
root@enterprise:~# grep -c "SSL" /var/log/vsftpd.log  
openssl s_client -starttls ftp -connect 127.0.0.1:21 </dev/null 2>/dev/null | grep -E "subject|issuer|Protocol"  
3  
subject=C=FR, O=Starfleet, CN=starfleet.lan  
issuer=C=FR, O=Starfleet, CN=Starfleet Root CA  
Protocol: TLSv1.3
```

Le FTP est bien chiffré



11. AUTHENTIFICATION WEB LDAP



12. HOLODECK BANC DE TEST

```
root@enterprise:~# apt install -y cockpit
mkdir -p /etc/systemd/system/cockpit.socket.d
cat > /etc/systemd/system/cockpit.socket.d/listen.conf <<'EOF'
[Socket]
ListenStream=
ListenStream=127.0.0.1:9090
EOF
cat > /etc/cockpit/cockpit.conf <<'EOF'
[WebService]
Origins = https://admin.starfleet.lan
ProtocolHeader = X-Forwarded-Proto
AllowUnencrypted = true
EOF
sed -i '/^root$/d' /etc/cockpit/disallowed-users
systemctl daemon-reload
systemctl restart cockpit.socket
ss -ltnp | grep 9090
```

admin.starfleet.lan - Cockpit

Administration web de la VM, derrière Nginx (HTTPS + LDAP), en écoute locale.

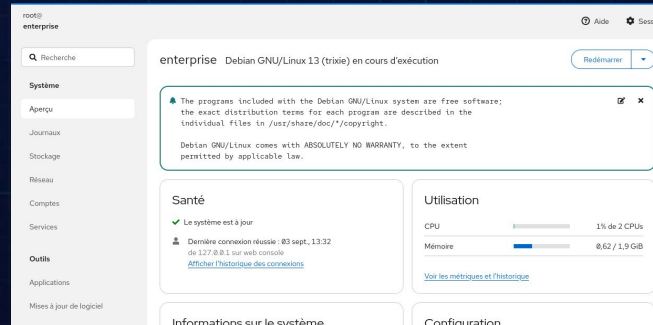
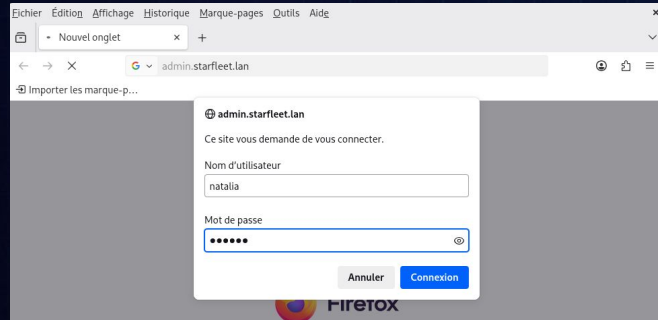
vscode.starfleet.lan - VS Code Server

code-server sur 127.0.0.1:8080, reverse proxy Nginx, auth LDAP.

Sauvegarde automatisée

backup-server.sh : réseau (ip, routes, ufw), services activés, paquets, /etc, /var/www, dump SQL, export LDAP → archive datée, cron 3 h.

13. BONUS



admin.starfleet.ian - Cockpit

Administration web de la VM, derrière Nginx (HTTPS + LDAP), en écoute locale.

vscode.starfleet.ian - VS Code Server

code-server sur 127.0.0.1:8080, reverse proxy Nginx, auth LDAP.

Sauvegarde automatisée

backup-server.sh : réseau (ip, routes, ufw), services activés, paquets, /etc, /var/www, dump SQL, export LDAP → archive datée, cron 3 h.

13. BONUS

```
root@enterprise:~# curl -fsSL https://code-server.dev/install.sh | sh
systemctl enable --now code-server@natalia
```

```
root@enterprise:~# sed -i 's/^auth: password/auth: none/' /home/natalia/.config/
code-server/config.yaml
sed -i '/^password:/d' /home/natalia/.config/code-server/config.yaml
systemctl restart code-server@natalia
cat /home/natalia/.config/code-server/config.yaml
bind-addr: 127.0.0.1:8080
auth: none
cert: false
```

→ écoute uniquement locale

```
root@enterprise:~# sed -e 's/admin.starfleet.lan/vscore.starfleet.lan/' -e 's/127.0.0.1:9090/127.0.0.1:8080/' /etc/nginx/conf.d/admin.conf > /etc/nginx/conf.d/vscore.conf
nginx -t && systemctl reload nginx
nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
nginx: configuration file /etc/nginx/nginx.conf test is successful
```

admin.starfleet.lan - Cockpit

Administration web de la VM, derrière Nginx (HTTPS + LDAP), en écoute locale.

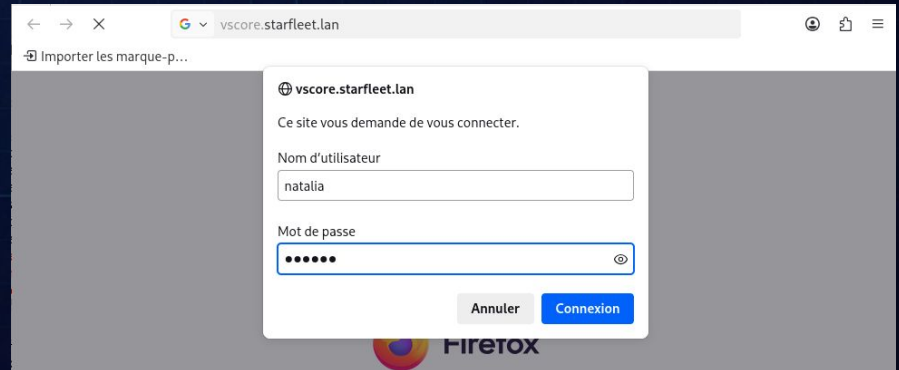
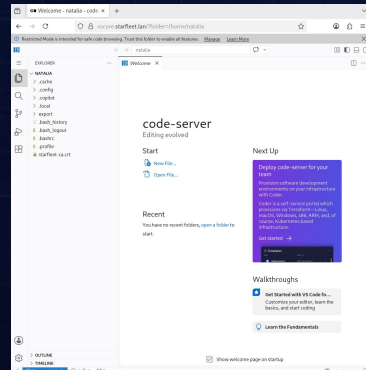
vscore.starfleet.lan - VS Code Server

code-server sur 127.0.0.1:8080, reverse proxy Nginx, auth LDAP.

Sauvegarde automatisée

backup-server.sh : réseau (ip, routes, ufw), services activés, paquets, /etc, /var/www, dump SQL, export LDAP → archive datée, cron 3 h.

13. BONUS



```
root@enterprise:~# cat > /usr/local/sbin/backup-server.sh <<'EOF'
#!/bin/bash
# Sauvegarde complete de la configuration du serveur enterprise
set -euo pipefail
DEST=/var/backups/holodeck
STAMP=$(date +%Y%m%d-%H%M)
WORK=$(mktemp -d)
mkdir -p "$DEST"

# Parametres reseau
ip -br a > "$WORK/ip-addr.txt"
ip route > "$WORK/ip-route.txt"
ufw status verbose > "$WORK/ufw-status.txt"

# Services et paquets installes
systemctl list-unit-files --state=enabled --no-pager > "$WORK/services-enabled.txt"
dpkg --get-selections > "$WORK/packages.txt"
cp -r /etc/apt/sources.list.d "$WORK/apt-sources"

# Configurations
tar czf "$WORK/etc.tar.gz" /etc 2>/dev/null || true
tar czf "$WORK/www.tar.gz" /var/www 2>/dev/null || true

# Donnees
mariadb-dump --all-databases --single-transaction > "$WORK/mariadb.sql"
slapcat -n 1 > "$WORK/ldap.ldif"

tar czf "$DEST/holodeck-$STAMP.tar.gz" -C "$WORK" .
rm -rf "$WORK"
find "$DEST" -name 'holodeck-*.tar.gz' -mtime +14 -delete
echo "Sauvegarde : $DEST/holodeck-$STAMP.tar.gz $(du -h "$DEST/holodeck-$STAMP.tar.gz" | cut -f1))"
EOF
chmod 750 /usr/local/sbin/backup-server.sh
```

admin.starfleet.lan - Cockpit

Administration web de la VM, derrière Nginx (HTTPS + LDAP), en écoute locale.

vscore.starfleet.lan - VS Code Server

code-server sur 127.0.0.1:8080, reverse proxy Nginx, auth LDAP.

Sauvegarde automatisée

backup-server.sh : réseau (ip, routes, ufw), services activés, paquets, /etc, /var/www, dump SQL, export LDAP → archive datée, cron 3 h

```
root@enterprise:~# echo '0 3 * * * root /usr/local/sbin/backup-server.sh >/dev/null 2>&1' > /etc/cron.d/holodeck-backup
```

13. BONUS

CONCLUSION

Toutes les contraintes du sujet sont couvertes

Pas de sudo · pare-feu minimal · HTTPS · dernières versions · PHP 7/8 · DHCP/DNS · starfleet.lan · FTPS chrooté · certificat unique · LDAP · VS Code Server et sauvegarde

Pistes d'amélioration

LDAPS (annuaire chiffré) · fail2ban (brute force) · sauvegarde externalisée hors VM · supervision (monitoring des services) · durcissement SSH par clés

DOCUMENTATION GITHUB

<https://github.com/natalia-giraldo/holodeck>

<https://github.com/anna-mamelli/holodeck>

<https://github.com/benahmed-ousseni/holodeck>

QUESTIONS ?

 **La Plateforme**

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